MYSX-256/512: A 256-bit Block Cipher with 512-bit Keys (v0.1-draft)

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2024

Abstract

We present MYSX-256/512, a 256-bit block cipher with 512-bit keys and 128 rounds. The design combines Feistel cells with Simon's round function, a Myszkowski byte transposition, word rotations, and an ARX key schedule. We provide the full specification, test vectors, and a self-analysis covering differential, linear, and related-key cryptanalysis. The differential analysis reports an observed upper bound on the best single-trail weight (114 at 14 rounds, extrapolating to the 128-round threshold at r≈15); this is not a proof of security and uses a Markov-independence assumption the design does not satisfy. The linear analysis attempted three models; none produced trails that verified empirically, so no linear bound is claimed. The key schedule was audited with 22 structural and empirical tests up to weight-6 with no persistent Δrk=0 pattern found. MYSX has not been externally audited. It is released as v0.1-draft for public review and must not be used to protect real data.

1. Introduction

MYSX-256/512 is an experimental block cipher. It is offered as a public proposal for cryptanalysis, not as a production primitive. The author makes no claim of security beyond the observed results reported here, and explicitly welcomes external review.

1.1. Security objectives (and non-objectives)

Model. MYSX is analyzed in the single-key, chosen-plaintext model for confidentiality, combined with an encrypt-then-MAC AEAD wrapper for integrity. The design does not claim security under related-key, chosen-ciphertext, or side-channel models; see Section 9.

Claimed security level. None formally. The paper reports observed upper bounds on single-trail weights and negative results on three linear-trail models. These are inputs to cryptanalysis, not proofs.

Non-objectives. MYSX does not claim:

- post-quantum security beyond the trivial Grover bound (2^{256} quantum iterations for a 512-bit key);
- resistance to related-key attacks beyond the heuristic coverage in Section 7;

- resistance to integral, impossible-differential, boomerang, or algebraic attacks (none analyzed);
- constant-time execution in JavaScript (side channels out of scope).

1.2. Design rationale

MYSX combines four building blocks:

- **Simon's F** `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`. Chosen for its proven diffusion/linear/ differential properties in Simon and its simplicity (no S-boxes, only AND/XOR/rotl). It does **not** inherit Simon's security proofs; the surrounding structure is different.
- **Feistel cells over 128-bit branches.** Four parallel Feistel cells per round, each acting on 32-bit word pairs.
- **Myszkowski transposition.** A fixed byte permutation derived from the keyword `MYSZKOWS`. The choice is aesthetic/didactic: it provides byte-level diffusion but no cryptographic argument beyond that. An optimized permutation would provide strictly better diffusion per round; see Section 9.
- **ARX key schedule.** A bijective Feistel 8+8 update with modular addition, XOR, and rotations. Chosen for simplicity and to avoid S-boxes; analyzed in Section 7.

The parameter choices (256-bit block, 512-bit key, 128 rounds) are not derived from a formal argument. They were chosen to:

- provide a large block (avoid birthday-bound issues of 64-bit blocks);
- provide a large key (Grover leaves 256-bit security);
- provide a large round count relative to Simon128/128 (68 rounds) as a precaution against unknown trails.

No claim is made that 128 rounds is sufficient. The round count is a parameter, not a security level.

1.3. Comparison with related designs

Design	Block	Key	Rounds	Structure	S-boxes
Simon128/128	128	128	68	Feistel	No
Speck128/256	128	256	34	ARX	No
ChaCha20	512	256	20	ARX	No
MYSX-256/512	256	512	128	Feistel cells + transposition	No

MYSX's closest sibling is Simon. The differences (block size, branch width, transposition, word rotations, non-linear key schedule) are substantial enough that Simon cryptanalysis does not transfer directly (see Section 8).

Performance. MYSX is estimated to run at a few MB/s in JavaScript, versus hundreds of MB/s for AES-GCM via WebCrypto (AES-NI). MYSX is not competitive on performance; it is a research primitive.

2. Specification

2.1. Parameters

Parameter	Value
Block size	256 bits (8 words of 32 bits)
Key size	512 bits (16 words of 32 bits)
Rounds	128
Structure	SPN-like: 4 parallel Feistel cells + byte transposition
Non-linearity	`F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`
Diffusion	Myszkowski transposition (keyword `MYSZKOWS`)
Rotation	`rotl(word, (r mod 31) + 1)` per word and round
Final whitening	XOR with 8 words from the key schedule

Structural note. MYSX is often described as a "2-branch Feistel" by analogy with Simon. Strictly, each round applies four independent Feistel cells over 32-bit word pairs, then a byte transposition over the full 32-byte state, then word rotations. It is more accurately classified as an SPN with Feistel cells. The "Feistel" label is a convenience.

2.2. Notation

- `rotl(x, n)`: circular left rotation of `x` by `n` bits.
- ∧: bitwise AND.
- ⊕: bitwise XOR.
- `s[r]`: state at round `r` (`s[0]` = plaintext, `s[128]` = ciphertext).
- `s[r][i]`: word `i` (0..7) of the state at round `r`.
- `rk[r][i]`: subkey `i` (0..7) of round `r`.
- `rkFinal[i]`: word `i` (0..7) of the final whitening.

2.3. Constants

```
CONST[0..7] = [
  0x243f6a88, 0x85a308d3, 0x13198a2e, 0x03707344,
  0xa4093822, 0x299f31d0, 0x082efa98, 0xec4e6c89
]
```

These are the fractional digits of π (same as Blowfish).

2.3.1. Myszkowski permutation

Keyword: `MYSZKOWS` (8 letters, S repeated at positions 3 and 8).

Alphabetical order: K(5), M(1), O(6), S(3,8), W(7), Y(2), Z(4).

The byte permutation `MYZ_PERM[0..31]` is constructed as:

```
MYZ_PERM = [
   4,12,20,28,                 # K (column 4)
   0, 8,16,24,                 # M (column 0)
   5,13,21,29,                 # O (column 5)
   2, 7,10,15,18,23,26,31,     # S,S (columns 2 and 7, read row-wise)
   6,14,22,30,                 # W (column 6)
   1, 9,17,25,                 # Y (column 1)
   3,11,19,27                  # Z (column 3)
]
```

The inverse permutation `MYZ_INV` satisfies `MYZ_INV[MYZ_PERM[i]] = i`.

Design note. The Myszkowski choice is didactic (classical transposition with a keyword) and adds no cryptographic argument. A byte permutation optimized for diffusion (e.g., a structured permutation over the 4×8 byte grid) would strictly dominate it. We keep Myszkowski in v0.1 because it is frozen; a future version may replace it.

2.4. Key schedule

2.4.1. Initialization

Input: key `K[0..63]` (64 bytes).

Initial state `s[0..15]` (16 words of 32 bits):

```
s[i] = big_endian_word(K[4i .. 4i+3])
```

2.4.2. Expansion

For `r = 0 .. 127`:

```
spread = (r + 1) * 0x9E3779B9  (mod 2^32)

# Round subkeys
for i in 0..7:
    rk[r][i] = s[i] ⊕ s[(i+5) mod 16] ⊕ s[(i+11) mod 16]
               ⊕ CONST[(r+i) mod 8] ⊕ spread

# For i = 0..3: fold fk into rk
for i in 0..3:
    fk = s[(i+8) mod 16] ⊕ CONST[(r+i+3) mod 8]
    rk[r][i] = rk[r][i] ⊕ fk

# State update (Feistel 8+8, bijective)
L = s[0..7]
R = s[8..15]
for i in 0..7:
    a = R[i]
    b = R[(i+1) mod 8]
    c = R[(i+3) mod 8]
```

```
    F = (rotl((a + b) mod 2^32, 7) ⊕ rotl(c, 13) ⊕ CONST[i] ⊕ spread)
    s[i]     = R[i]
    s[i + 8] = L[i] ⊕ F
```

2.4.3. Final whitening

After the 128 rounds:

```
for i in 0..7:
    rkFinal[i] = s[i] ⊕ s[(i+8) mod 16] ⊕ CONST[(128+i) mod 8]
```

2.5. Round function

Each round `r` transforms state `s[r]` into `s[r+1]` in four steps:

Step 1. AddRoundKey.

```
for i in 0..7:
    s[r][i] = s[r][i] ⊕ rk[r][i]
```

Step 2. Feistel cells. For `p = 0..3`:

```
L = s[r][p]
R = s[r][p + 4]
s_mid[p]     = R
s_mid[p + 4] = L ⊕ F(R)
```

where `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`.

Step 3. Myszkowski transposition. Convert `s_mid[0..7]` to bytes (big-endian per word):

```
bytes[4i .. 4i+3] = big_endian(s_mid[i])
permuted[j] = bytes[MYZ_PERM[j]]  for j = 0..31
s_perm[i] = big_endian_word(permuted[4i .. 4i+3])
```

Step 4. Word rotation.

```
shift = (r mod 31) + 1
for i in 0..7:
    s[r+1][i] = rotl(s_perm[i], shift)
```

2.6. Encryption

```
s[0] = plaintext  (as 8 big-endian words)
for r = 0..127:
    apply round r (Section 2.5)
for i = 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
ciphertext = s[128]  (as 32 big-endian bytes)
```

2.7. Decryption

Inverse of encryption:

```
s[128] = ciphertext
for i = 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
for r = 127..0:
    apply inverse round r (inverse of Section 2.5, reverse order)
plaintext = s[0]
```

The inverse round consists of:

1. Inverse rotation: `rotl(word, 32 - shift)`.
2. Inverse transposition: `bytes[MYZ_PERM[j]] = permuted[j]`.
3. Inverse Feistel: `R = newL`, `L = newR ⊕ F(R)`.
4. XOR with `rk[r][i]`.

2.8. Mode of operation (CTR)

32-byte counter:

```
counter[0..15]  = nonce (16 bytes)
counter[16..31] = 0  (128 bits, big-endian)
```

For each 32-byte plaintext block:

```
keystream = MYSX_encrypt(counter)
ciphertext_block = plaintext_block ⊕ keystream
counter[16..31] += 1  (big-endian)
```

2.9. AEAD (encrypt-then-MAC)

Packet: `VERSION(1) ‖ NONCE(16) ‖ CT(n) ‖ TAG(32)`

MAC input: `VERSION ‖ NONCE ‖ AAD_LEN(4 BE) ‖ AAD ‖ CT_LEN(4 BE) ‖ CT`

TAG: `HMAC-SHA256(macKey, MAC input)`

Verification: the decryptor verifies the TAG before decrypting.

2.10. Key derivation

From a random secret (HKDF-SHA256).

```
encKey = HKDF-SHA256(secret, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(secret, salt, info = context ‖ ":mac", L = 32)
```

From a password (PBKDF2-SHA256 + HKDF).

```
prk    = PBKDF2-SHA256(password, salt, iterations = 600000, L = 32)
encKey = HKDF-SHA256(prk, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(prk, salt, info = context ‖ ":mac", L = 32)
```

2.11. Test vectors

See `KAT.json` (public). Two digests are documented in `README.md`:

- `KAT.json` public digest: covers the full JSON (block + AEAD + KDF).
- `KNOWN_KAT_DIGEST` (internal): covers 16 blocks with `key[i] = i`, used as fast regression in `mysx.test.js`.

3. Differential cryptanalysis

3.1. Model

Tool: `mysx_milp.py` (PuLP + CBC/HiGHS). AND model: `a = Δx ∨ Δy`, which **overestimates** the cost of an active AND (counts a cost-1 event whenever at least one input is active). This is a heuristic; it is not a proof that the model represents all differential transitions. Specifically:

- The Markov-independence assumption does not hold: the two rotated inputs of F come from the same word `x`.
- We ran a single input difference at r=12–14 (w2b30). Multi-bit input differences may yield lower weights.
- `gapRel=0.2` and `timeLimit=300` return feasible solutions (upper bounds), not certified optima.

3.2. Observed upper bounds

Rounds	Input	Weight (upper bound)	Single-trail prob.
5	w1b24	2	2^{-2}
6	w1b24	6	2^{-6}
7	w2b30	10	2^{-10}
8	w2b30	18	2^{-18}
12	w2b30	68	2^{-68}
14	w2b30	114	2^{-114}

Extrapolation: the 128-bit threshold is reached at r≈15. **This is an extrapolation, not a proof.** It tells us what the model produces for the input we tried; it does not establish the best trail over all inputs, nor that clustering does not lower the effective security.

3.3. Empirical check (r=4, w0b0)

N=2,000 samples; match rate 0.5–1.15% (prob ≈ 2^{-7}). The model predicts 2^{-8} for this input; the factor-2 gap is consistent with differential clustering. This is a sanity check, not a proof of the extrapolation.

3.4. What this section does not establish

- A lower bound on the best single-trail weight (requires gap=0 and multi-bit inputs).
- Resistance to differential clustering (requires modeling multiple trails with the same output difference).
- Resistance to related-key differentials.

4. Linear cryptanalysis

4.1. Three models, no bound

We tried three linear models. **None produced trails that verified empirically.** The honest conclusion is that no linear bound is established for MYSX.

4.1.1. Hybrid model (`act = α ∨ β ∨ γ`, no `γ ≥ α`)

This model produced growing weights (8 at r=3, 18 at r=4, 86 at r=6, 434 at r=10). Empirical verification at r=4 (weight 8, expected correlation $2^{-8} = 0.0039$) gave correlation -2.4×10^{-4} over 2,000,000 samples, at 0.53σ from zero. **The predicted trail does not exist.**

Diagnosis: the model allowed `(α,β,γ) = (1,0,0)`, which has correlation 0 in the real AND (not 1/2). The trails found were artifacts.

4.1.2. Strict model (`x = y = z = w` in XOR)

This model returned INFEASIBLE in all rounds 1–10 for mask w0b0. The model is too restrictive: it forbids valid trails with correlation 1 (e.g., the trivial r=1 trail with weight 0).

4.1.3. Relaxed model with correct AND (`x+y+z-w=2u, γ = α OR β`)

Growing weights (8 at r=3, 18 at r=4, 86 at r=6, 434 at r=10). Empirical verification at r=3 (weight 8) gave correlation -2.4×10^{-4} over 5,000,000 samples, at 0.53σ from zero. **The predicted trail does not exist.**

Diagnosis: model artifact, or destructive clustering (multiple trails with opposite signs cancelling). **We do not know which.** Either way, the conclusion is the same: we have no verified linear trail, and therefore no linear bound.

4.2. Why this matters

The best published attacks against Simon128/128 are linear (hull attacks covering ~49/68 rounds, complexity $\sim 2^{127}$). The linear side is therefore the most likely attack surface for MYSX. **A negative result on three models is not a security claim.** It is an open question.

To establish a linear bound for MYSX, future work would need:

1. **Sign tracking** (±1/2 per active AND), which the current MILP does not model.
2. **Linear hull analysis** (clustering of many trails with the same mask).
3. **A SAT/MILP model with XOR correlations < 1**.

None of these is done here. **MYSX's linear security is unknown.**

5. Related-key cryptanalysis

Tool: `keyschedule_test.js` (basic) + `keyschedule_test_extended.js` (22 tests). Coverage: exhaustive weight-1 (512 bits), exhaustive weight-2 (130,816 pairs), weight-3 "same Δ" (1,680 trials), beam search up to weight-6 (beam 50), 5 base keys, structured patterns, cycles, weak keys.

Results (summary):

- No persistent `Δrk[r]=0` for r≥1 in any test.
- Beam search score: 139 (ideal 128).
- No weak keys, cycles, or fixed points.
- Slow diffusion at r=1–4 (expected: one Feistel round per cipher iteration).
- Worst bit (minimum weight at r=1): bit 160 (word 5, bit 0), 1-bit difference. Consistent across 5 base keys (structural, not noise).

Limitations. These are heuristic tests, not a proof. We did not run a targeted MILP/SAT related-key model (~650k variables); adaptive related keys, biclique attacks, and algebraic attacks on the key schedule are not covered. **MYSX's related-key security is unknown.**

6. Generic attacks

Attack	Complexity	Verdict
Classical brute force	2^{512} operations	Infeasible
Grover (quantum)	2^{256} iterations	Infeasible with foreseeable technology
Block collisions (birthday)	2^{128} blocks	Infeasible

Note on Grover. With a 512-bit key, Grover leaves 256 bits of effective security. With 1,000,000 logical qubits at 1 ns/iteration, the time would be $\sim 10^{60}$ years. This is a theoretical estimate; no post-quantum analysis of the cipher beyond Grover has been done.

Note on the 512-bit key. HKDF-SHA256 produces at most a 256-bit pseudorandom key. If the master secret has 256 bits of entropy, the effective security is 256 bits, not 512. The "key size" of MYSX is not the security level of the system.

7. Simon cryptanalysis: what transfers and what does not

MYSX uses Simon's `F`. It is **not** Simon. The differences:

- Block/branch size: 256-bit block with 128-bit branches (Simon uses `n`-bit branches).
- Four parallel `F` per round (Simon uses 1).
- Myszkowski byte transposition (Simon has none).
- Round-dependent word rotations (Simon has none).
- Non-linear ARX key schedule (Simon's is linear).
- 128 rounds vs 68.

What transfers. The linear and differential distinguishers on Simon's `F` are properties of the function, not of the key schedule. Any distinguisher on F may in principle be embedded in MYSX's structure. The Simon-specific attacks on round-reduced Simon do not transfer directly because the surrounding state, diffusion, and round count are different, but the F-level biases are the natural starting point for attacking MYSX.

What does not transfer. Key recovery from Simon's linear key schedule does not transfer: MYSX's key schedule is non-linear.

Relevant literature.

- Kölbl, Leander, Tiessen, "Observations on the SIMON block cipher family", CRYPTO 2015.
- Biryukov, Roy, Velichkov, "Differential analysis of block ciphers SIMON and SPECK", CRYPTO 2015.
- Chen, Wang, "Improved linear hull cryptanalysis of round-reduced versions of SIMON", ePrint 2014/973.

None of these results is transferred to MYSX in this paper.

8. Analysis limitations

- **Differential**: upper bounds on single-trail for one input at r=12–14; no lower bounds; no clustering.
- **Linear**: no verified trail in three models; no bound.
- **Related-key**: heuristic tests up to weight-6; no targeted MILP.
- **Integral, impossible, boomerang, algebraic**: not analyzed.
- **No independent public cryptanalysis.**
- **Empirical verification**: consistent with the model at r=4 (differential) and inconsistent at r=3–4 (linear).
- **Functional tests** (`mysx.test.js`, 16/16 PASS) cover round-trip, AEAD negatives, nonce policy, PBKDF2, avalanche. They do not prove security.

9. Engineering notes

9.1. API surface

- Production API: `aeadEncrypt`, `aeadDecrypt`, `deriveAeadKeys`, `deriveAeadKeysFromPassword`, `NonceGuard`.
- Test API: gated by environment variable `MYSX_ENABLE_TEST=1`. Not activatable from code (`globalThis`), only from the process environment.

- CTR (`_ctrXor`) is not exported from the production API.

9.2. NonceGuard

`NonceGuard` is a best-effort defense against nonce reuse. It:

- counts messages per key (default 2^{32});
- detects recent nonce reuse (window of 1024 per key);
- **does not persist across restarts**;
- **does not share state across processes or devices**;
- caps the number of tracked keys (default 1024) to avoid unbounded memory growth;
- provides optional persistence callbacks (`onSave`, `onLoad`) for applications that need it.

It is not a substitute for key lifecycle management. If the application reuses `(key, nonce)` across restarts, CTR confidentiality is broken. See `mysx.prod.js` for the exact contract.

9.3. aeadEncrypt size validation

`aeadEncrypt` validates plaintext size **before** allocating the CTR output buffer, and enforces a configurable `maxPlaintextBytes` limit (default 2^{32}). Applications that require larger limits should raise the limit explicitly and manage memory accordingly.

9.4. KAT digests

Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact that a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**; the internal digest is only a self-check.

10. Verdict

MYSX-256/512 v0.1-draft is an experimental proposal, not a secure cipher. The differential analysis reports upper bounds on single-trail weights, not a security proof. The linear analysis is a negative result on three models, not a bound. The related-key analysis is heuristic. No integral, impossible-differential, boomerang, or algebraic analysis has been done. The cipher has not been externally audited.

Do not use MYSX to protect real data. For production: **AES-256-GCM** or **XChaCha20-Poly1305**.

11. Reproducibility

File	Purpose
`SPEC.md`	Formal specification
`PAPER.md`	This document
`KAT.json`	Test vectors (block, AEAD, KDF)
`generate_kat.py`	KAT generator
`mysx.prod.js`	Implementation (production API)
`mysx.test.js`	Functional tests
`mysx_milp.py`	Differential MILP
`mysx_linear_milp.py`	Linear MILP (three models)
`keyschedule_test.js`	Basic key schedule audit
`keyschedule_test_extended.js`	Exhaustive audit (22 tests)
`check_trail.js`	Empirical differential trail at r=4
`check_linear_trail2.js`	Empirical linear trail at r=3

Fixed seed in tests (`0xC0FFEE`). MILPs are deterministic given the solver and `gapRel`.

12. Final warning

MYSX has not been externally audited. This paper is the author's self-analysis, not a certification. No result here replaces an independent cryptographic review.

Do not use MYSX to protect real data.